

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 What is computer vision? Explain how it complements NLP? Give applications and future scope of Computer Vision? (CO5)
- Q.24 Define ANN? Classify and list characteristics of Artificial Neural Network? Give 5 applications of ANN? (CO3)
- Q.25 The malevolent utilization of AI is an unfortunate reality within our society. Elaborate on this phenomenon with pertinent examples. (CO1)

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Roll No.

2nd Sem / Artificial Intelligence & Machine Learning

Subject : Fundamentals of Artificial Intelligence

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which of the following is NOT a fundamental challenge faced by AI? (CO1)
- a) Realism
 - b) Scalability
 - c) Common Sense Reasoning
 - d) User Friendliness
- Q.2 AI-powered virtual assistants in smart homes are most likely used for: (CO1)
- a) Treatment of Pets
 - b) Controlling lighting
 - c) Performing financial transactions
 - d) Assisting with medical diagnosis
- Q.3 What are the security-relevant properties of AI that are crucial for preventing and mitigating malicious use? (CO2)

- a) Color and shape recognition
 - b) Learning speed and memory capacity
 - c) Processing and energy efficiency
 - d) Accountability and robustness
- Q.4 Least suitable language for AI programming is:
(CO2)

- a) C++ b) JAVA
 - c) COBOL d) MATLAB
- Q.5 Which of the following is a potential concern associated with deep fake technology? (CO1)
- a) Spreading misinformation
 - b) Manipulating digital transactions
 - c) Augmenting medical diagnostics
 - d) Enhancing data encryption
- Q.6 Which component of NLP involves the analysis of the meaning of words and sentences? (CO5)
- a) Semantic analysis
 - b) Syntax analysis
 - c) Morphological analysis
 - d) Pragmatic analysis

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Expand DL? (CO4)
- Q.8 Define Supervised Learning? (CO3)
- Q.9 Define Linear Activation function of ANN? (CO3)

- Q.10 Give one application of AI in healthcare sector? (CO2)
- Q.11 Define deep fake? (CO1)
- Q.12 What does the term Artificial Intelligence indicate? (CO1)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 List various categories of AI? Also give their distinguish features? (CO1)
- Q.14 Compare and contrast human intelligence with machine intelligence? (CO1)
- Q.15 In your opinion, what is the predominant challenge encountered by artificial intelligence (AI)? Why? (CO2)
- Q.16 Is it conceivable for AI to genuinely attain common-sense knowledge? Justify your conclusion? (CO2)
- Q.17 What does common-sense reasoning entail? (CO2)
- Q.18 Write a note on machine learning? (CO4)
- Q.19 Differentiate between Feed forward and Feedback Neural Network? (CO3)
- Q.20 Explain various challenges of Natural Language Processing? (CO5)
- Q.21 What is Deep Learning? Explain? (CO4)
- Q.22 Discuss the application of AI in the realm of cyber security? (CO1)